

PLANETVORTEX

PLANETVORTEX — Lemma E — the mass-ladder deficit

The theorem edition of the planetary gravity-geometry bench — mini-repository planetvortex/ of the TRIVORTEX framework, version 1.0.0. This dossier binds one statement to its registered protocol; the full science lives in the big monograph (docs/monograph/).

Statement

The gravity ladder $\delta_i = \log_{10}(G M_i / G M_{\oplus})$ carried by the seven stations spans -1.26 (Mercury) to $+2.50$ (Jupiter). Summed over the seven Fano lines it ranges from 0.008 (Mercury+Mars+Neptune) to 6.715 (Jupiter+Saturn+Neptune) — a spread of 6.71 dex. Dynamically, a mild 10% gravity-weighted circulation ladder deforms the heptagon lattice by 0.298 within one rotation, while the equal-circulation ring stays rigid to 9.8×10^{-15} .

Proof

This is a recorded diagnostic lemma: the statement binds the committed NASA register (a static sum) and the Kirchhoff flow (a dynamic drift) to the mass-blind figure. The protocol records both registers verbatim — the line sums per line and the two radius-drift records — with no pass/fail role: the deficit is the bench’s honest headline, not a failure. The mean-motion ladder (684.1 from Mercury to Neptune) against the chord ladder (2.247) is recorded in the same block. ▫

Protocol binding

item	value
stage	P7
recorded registers	$6.71 \text{ dex} / 0.298 \text{ vs } 9.8 \times 10^{-15}$
protocol file	results/protocols/ (default preset)
test guard	tests/ re-derives every preset-independent register

Cross-references

- the parent framework: Theorem 3.1 and the three-vortex anchor;
- the sibling bench polyvortex: the W2/W4 registers at $N=7$ — the same heptagon, cross-pinned bit-for-bit;

- the big monograph: Sections 3 (Theorem A), 6 (Theorem C), 7 (Theorem B), 4 (Lemma D), 8 (Lemma E);
- the formalization queue: roadmap item V4 (Coq / Lean 4).